

ActInf Livestream #006.2 ~ "A tale of two densities: active inference is ena

Livestream | (2020) | 1:10:11

in behavioral ecology and
pathology and evolutionary biology with
with models that sort of
[Music]
having some sort of empirical conflict
cool anyone else want to share something
they're working on
or i can also just raise the second
warm-up question
just in case people want to answer that
one which is what is something that
you'd like to have resolved by the end
of today's discussion
whether it's a thought about the paper
specifically or
something general um sure
fellow jitzer i'm not sure that is
shannon go ahead
hey um i wonder
whether the uh like structural
representation
that we might be defending later today
um whether that structure could just be
a set of attractor
dynamics like is that enough to be a
structure
um so i think
alex might be raising his hand to have
something to say
yep alex oh yeah i mean i was gonna
speak anyway
i mean i would say yes to that question
probably
um it's not just my opinion but um i was
gonna answer

both questions i guess so um apart from
trying to figure out the free energy
principle still uh lots of conversations
with people i've been trying to um
i've been working up to trying to
implement a variational autoencoder to
figure out how that actually works
so um been doing a lot of sort of more
technical work um something i'd like to
have resolved is just
to figure out exactly where um
in activism or the version of an
activism that's defended in this paper
is inconsistent with representationalism
if it is um
so like it's not i don't need there to
be residual disagreement but
i just want to figure it out
sounds good any last thoughts on these
warm-ups before we
jump into it we have a lot of very
interesting slides kate go for it
yeah i guess the biggest thing that i
want to resolve by the end of the
session is i'm still not
really clear on the status of the
generative model because at some points
it sounds as if
that's something that's almost causal
that it's controlling the animal's
behavior or
the organism or the system's behavior
and at other points it's
a model it sounds more like it's a model
that we use to describe the systems
dynamics rather than
something that is including beliefs or
being causal and that's something i'm

still not entirely clear on so i think
that's
a big thing cool
all really interesting topics thanks for
sharing and we're gonna
jump into it because a lot of these are
on the slides ahead
so the paper that we're going to be
coming back to today is
a tale of two densities active inference
is an active inference by
ramstead kirchhoff and fristen and just
to rehearse the goal of this paper
they aimed to clarify how best to
interpret some of the central constructs
that underwrite the free energy
principle
and its corollary active inference in
theoretical neuroscience and biology
namely the role that the generative
models and the recognition densities
play in this theory
aiming to unify life and mind and so
last week we talked about what these two
densities are
and we talked about the tail that
integrates the densities
we also walk through the road map how
they build from a to z
by talking about statistical models and
representations
introducing active inference and then
combining that with an
active inference in this sort of
inactivism 2.0
framework before building towards
multi-disciplinary research heuristics
for cognitive science

so that was last week let's get into the
new and exciting parts
so this was one of the feedback comments
from a participant
last week and what they wrote was a key
learning goal for me in building my
understanding of active inference is to
develop
intuitive examples for the critical
concepts that can act as the basis for
abstraction later on
it seems to me like there is probably
something like a logical sequence in
which this should be done but i don't
necessarily always know which concepts
and ideas from active inference to
prioritize
as an example coming from a cognitive
linguistics background i tend to think
of beliefs as either causal reasoning
or the underlying sensory motor
simulations that connect to linguistic
concepts
i've been trying to find new intuitive
examples that can help me to understand
beliefs as probability distributions as
clearly this is the key for making
progress in understanding active
inference
so my question for all of you is what is
a system
or a metaphor that helps you understand
active inference
and why and while people are raising
their hand on the bottom we have
just a few of the systems that we've
been discussing over the last weeks
we have a brain the technical and the

artistic parts of the brain
we have insect colonies we have dance
and embodiment and movements and then on
the right i guess it reflects
something like the economy or changes in
dynamical systems but
what is a system that helps you
understand
through specifics something about active
inference
any thoughts on this
i mean i hate to be that guy but like i
started
understanding active inference when i
went into the math
uh like it's it's a formal theory and i
i think we can actually be misled by
uh metaphor uh
yeah uh some of my some of my
reservations about the philosophical
work that's been done on active
inference is that
it it operates on the basis of you know
um kind of unformal
intuitive conceptual understandings
rather than trying to
see what the math that says so i hate to
be that guy though
no that's perfect we'll go to alex
fiatkin and then
to kate go ahead alex
yeah for me most important changed
my views was uh taking seriously
approach from like for process
and understanding uh
concepts about states and steady states
of possible processes and the later on
all this concept with markov blanket as

a border
and underlying structure like
generative models in links with
cybernetics it's bringing me
some kind of top-level picture and now
i'm trying to
focus on different parts to have more
detailed understanding
cool kate and then mel
yeah i was also going to say cybernetics
i think i only really started to
understand what was going on once i
started seeing the links with like ashby
and accounts of survival as stability
um that was when things started to make
sense for me um
and there's also just like a nice
metaphor i quite liked and i think it's
brilliant
it's a paper by ellie bruneberg i think
with the union keemstar
right with the crooked scientist
metaphor the way the scientist is trying
to
obtain the result that it's looking for
so it has to have some sort of
latch on the world in order to bring
about that result but it's still trying
to obtain
the result that it wants in the first
place rather than passively
find out what will happen i thought it
was quite a nice metaphor
so do you think they are helpful cool
mel
so i remember i'm i'm remembering um
my first sort of aha moment with the fpp
and active entrance

um i was i was explaining this stuff
to my friend who works on on similar
things but in a very different framework
um in the mit media lab and he had like
papers
strewn everywhere scribbling all over
them and
my realization in explaining this
framework to him was that
um that there is
fundamentally at the at the lowest level
at the at the level of the simplest sort
of systems
or at their kind of lowest scale
right um no difference between
belief updating and action so
a a change in the system is a change in
the system
right and that is both
a belief update and an action right
um and once you build in
further levels of of hierarchical
organization and complexity you get this
strong differentiation between them but
at the lowest level of this the same
thing

[Music]

cool and kate did that
okay had you raised your hand there and
also i'm just
going to yeah alex kiefer go ahead
oh um yeah i mean i was gonna
that's i like that point that at least
at the lowest level we'll leave updating
an action or kind of
same thing i guess um that's one reason
that's one reason that i'm skeptical
that we need to explicitly i think

implicitly

more traditional bayesian brain stories

kind of

are built on that claim as well but um i

was going to say what got me into what

got me into active inference also

sort of a technical point was just that

you really can't you can't optimize the

thing you can't minimize surprisal

just by ingesting adjusting and

at least not okay the surprise

depends in part on the sensory

observations right so you can't minimize

that

just by changing your generative model

um you have to

change the observations so that's that's

the technical thing that got me into it

um cool and yeah

thanks everyone for being flexible and

fun on the tech

maxwell i really agree that the metaphor

as always with metaphors it's a question

about

compression of semantics and meaning and

at the end of the day we're talking

about

something specific and so we can think

about different systems

that help us understand how active

inference is deployed in the real world

but often metaphors will take off

take us off the mark uh because it

removes us from the real

underlying details so let's return and

the details are really where the devil's

at

uh in this case uh i think

but yep i i agree hopefully in the end
with the figures we're gonna be able to
look
at those details really specifically and
i think that the way that figure 2 3 and
4
lead into each other helped me
understand a lot about the system
so just to take another look at these
two
cities the two cities that are being
linked the densities
on the left we have the bayesian
structural representationalist
and alex freely added any details here
but bayesian models move between the
data parameters the observations and
hyper parameters or higher order
parameters and
from the data to the hyper parameters we
call it a recognition model
and to go from hyper parameters to a set
of data or observations
that is a generative model and the
outcome is a statistical convergence of
a multi-level model that represents
structures of the world
for example through an expectation
maximization scheme
we can contrast this with the inactivist
school of thought
which has linked rather than data and
hyper data
but rather has focused on the world and
agents and how they're linked through
perception
and action so a lot more adjacencies to
areas

like niche construction and ecological psychology
and we remember hearing from maxwell last week about how it was the desire to mathematically rigorize some of the inactivism that led him to the bayesian approach and let's now think about how these two densities are linked with a little bit more feedback from one of our participants they wrote the domino metaphor finally makes sense the physical dominoes are the physical state of the system corresponding to the recognition model and the falling of the dominoes is like the process or the dynamics of the system reflecting the generative model so here we've kind of combined these two different ways of thinking and we're looking at the relationship of the world and the agent or the system and the system's surroundings and directly went ahead and combined perception with the recognition model and action with the generative model and then also just to combine that with what kate was just talking about with cybernetics this is a quote from the paper that says on this view active inference can be read as a new take on the good regulator theorem proposed by control and rosh ashby in 1970. active inference tells us about the

relation between a control system
the generative model with priors over
action policies
and a system being controlled the
organism and its adaptive behavior
the actual actions undertaken in and
part of the world
so what exactly does this clarify
about active inference
or what do people think about this
metaphor um or at least way to think
about how
if not necessarily a metaphor how would
we apply active inference to this
system what are we getting at here with
these similarities and differences
between the two
models
sure well mel
the the good regulator theorem the idea
is that um
any good control structure for some
larger system has to
be a model of that system it has to
contain
in it all of the the key variables
for that system and basically what this
says
to me about active inference is that um
in order for a system
in an environment to be able to
adequately
react to whatever kinds of perturbations
are happening in the environment
it has to contain all of the key
existence i call them existential
variables and
the the variables in the environment

that would lead to the system
continuing to exist or not continuing to
place
cool and so how exactly i'm just curious
um does the system come to embody all
the key variables of the outside world
like
don't the models of the organism reflect
a simplified
version of the outside world otherwise
there kind of is a abortion story about
the map that represents the territory
exactly
so how does a good regulator arise in
the context of an environment like that
um shannon and then maxwell
well i don't think the key is to be an
exact
like imitative model of the world or an
exact map even of the world
but just a model that is good enough to
enable you to act and
adjust in the world
cool maxwell and then mel
yeah i mean uh so formally speaking
there's a difference between
external states and hidden states um
and i mean when everything is going well
uh the external states that are modeled
by the system coincide with
the hidden states that are actually out
there
um and this is more of this it's not
i'm not i'm not sure if it's guaranteed
but
it it is strongly implied by the fact
that we're minimizing
uh variational free energy uh

that you know it if we're if we're
not generating a lot of
free energy then uh the external beliefs
that we have
about the environment tend to reflect uh
you know the causal structure of that
environment
but i mean manuel baltieri and chris
buckley have done some interesting work
showing that actually
there isn't a necessary kind of
entailment relation
um you can have uh
you can have the the system can be
acting
that don't actually exist and i mean
this makes sense also
just evolutionarily uh perceptual
systems don't track truth they enable
adaptive
uh behavioral loops so if you're a prey
item like a bunny
it's adaptive to generate a lot of false
positives
uh compared to you know allowing for
false negatives and in terms of like
predator detection
it's it's more advantageous to you know
make a few
less costly mistakes than make a big one
so uh
i mean yellow breenenberg and eric riel
make make that point in the
paper that kate was mentioning just
earlier but uh
yeah the it doesn't actually have to
hook up
to anything in the real world though

usually it will end up
doing so cool mel and sasha
yeah so the the key term in what i said
is key actually um so the key variables
right
that's that's just what keeps the system
alive and we can imagine
that we first get a system on the scene
that
needs just basically one parameter
in its environment to be correct in
order to continue to exist as a system
and then we build up from there to
systems that are progressively
more and more complex and need
progressively more and more
complex sort of environmental scenarios
to survive
and then they themselves in order to
survive in those
more complex regimes need to build in
complexity
right in in terms of their their the
models
of the world that they enact right and i
think it says something
interesting about um how we
conceptualize a
cognition and the onset of cognitive
complexity
um as sort of um and this is
peter godfrey smith's line on subject is
is it's a response to a complexifying
environmental circumstance right
cool sasha then alex keeper
um yeah uh one key phrase that really
helped me better
understand um active inference

um is uh good enough and that's what
really
um kind of put it all together that um
while we're
going about and reducing uncertainty
about our system
it just has to be good enough to make
the next action
or in the evolutionary sense to survive
and
that metaphor has really helped me go
through all this
um all these high level concepts so i
thank you for mentioning that
alex kiefer then alex fiatkin thanks um
yeah i just wanted to just note that
there's there's you don't need
to contrast accuracy or you know
answerable to the truth-ness with um
uh enabling adaptive behavior uh as as
like a dichotomy
um i'm not sure if that's what was
intended or not but this one reason i've
had trouble getting into
understanding the sort of
anti-representationalist
viewpoint is that um you know structural
similarity is a matter of degree
accuracy is a matter of degree in this
sense so um
i mean yes your hidden state
representation doesn't have to map
completely
accurately onto reality it doesn't have
to capture every detail but if it
doesn't do that to any degree
you're screwed so you know maybe we need
systematically sort of simpler or

biased representations in order to do
the job but i think there's still a
implied relationship to the truth
there cool alex
yeah thanks uh i want to change
the system's level of consideration of
possible application of active inference
and for me interesting
on a personal level in terms of
day phenomenology and especially
professional and or working day
phenomenology
how it works what is a generative model
and like an example
and what i won't discuss if i get it
correctly or
how it's good possible develop to think
about it
but if for example some doctor
he know different disciplines
and with different ontologies
and when he met the patient he
started his actions by his generative
model which activates with
recognition models to serve
as a doctor depends on exact situation
and exact case
and if it's so from for example from
another
level from team level if person
behave professionally in some discipline
it
became like a perception model
for a generative model of the team
and what does it mean in terms of
learning and education of team members
because
if we can how

some in some way to link recognition
models with ontologies
and starting to work with it more again
cybernetically it's possible could be
very interesting
at least for me cool agreed the the
ontologies that how we think about the
world
definitely influences our perception
and sasha's do you have something else
otherwise okay shannon go ahead
um i was just going back to alex
keefer's point that
your model like there has to be some
truth relationship with the world and if
we're going
to the dominoes and the process needs to
be that the dominoes fall to make some
cool pattern
if there's an interruption like so now
this domino can't reach
the next domino like that's akin to a
disruption and
how truthy the model is of the world
and it disrupts the action that you're
able to take or it disrupts the
um you die instead of being able to
survive
because your model isn't accurate at
that stage for the processes to continue
happening to
do the next action or to survive
cool cool discussion on this domino
system let's talk about counter factuals
and also return
to some of the great questions alejandra
raised in previous
weeks this is from another participant's

feedback

they wrote what i'd like to clarify if

possible is the following

am i right in understanding the

generative model as a control system

that is dynamically instantiated

between an organism and an affordance in

the context

and the recognition density as a

relation between the sensory inputs and

the possibilities for action policies

encoded in the organism

if so how does this work when action is

instantiated by counterfactual

forms of cognition or simulation i.e

those that do not rely on affordances in

the current external context

and so on the bottom right there's a

younger person

looking into the mirror or maybe vice

versa and seeing an older

person and this is reflecting how not

only is there a deep temporal

aspect to our action in the current

moment

meaning that there's often affordances

that aren't available to us at the

second that we're

actually making the decision but also

that we think

about future possible affordances that

don't exist based upon the sensory data

that we're getting

or do they and then on the right side i

have a few bacteria

because this is like an example of a

system

that we might not think of as having a

simulation or
counterfactual based mechanism of action
often bacterial behavior can be
summarized quite concisely by just the
short term
gradients that it's ascending or
descending so
for anyone to raise their hand and chime
in
how how do counter factuals play into
this and how do we make sense of
affordances
that aren't in a current ecosystem but
are still important for behavior maxwell
well i just want to comment on the first
part of that question
um so i mean i think it's a bit simpler
than that in terms
of the relationship between the
recognition and the generative
models or densities um so your
recognition model is just
your post your posterior the all of your
posteriors basically all of the
all of your posterior estimates over
states
and precision it's like my best guess
as to you know what the value of all
these states
are um and
i mean you can think of that as uh the
system's current
physical state uh so under under the act
of inference
um formalism the current physical state
of the organism encodes uh
basically the the specific parameters
of these uh beliefs over external states

the
the actual physical states of the system
encode these parameters which
so yeah you embody a best guess
the the generative model is the point of
reference for the free energy gradients
so you can think of one as like you know
uh
the inference model and the other is the
control
model in the sense that the inference
model of the recognition density is sort
of like my best guess right now
and the generative model is is
essentially you know a model of what my
phenotypic preferences are
and uh it's the tandem between the two
that kind of gets you behavior
as in um the generative model provides
the point
uh the point of reference uh for the
definition of the variational free
energy gradients i.e
how close am i um to
my you know desired sensory distribution
and the recognition model tells you
where you are effectively
so it's like where i am where i want to
be and the free energy
is sort of the difference between the
two effectively um
so i mean the first part of that
question
yes you're right in understanding the
generative model as a control system
um the recognition density is just the
best guess
it's the posterior uh so i mean it yes

it is harnessing these relations that
are being described here
uh but that's precisely because it's a
posterior estimate
of uh you know states and precisions and
all that
cool and one more thought on that is
for an agent who knows about for example
how hammers and nails are related
through their internal models they might
look at hammers and nails and pieces of
wood
and think you know later on i could
build a house with this
and so it's actually a future affordance
but it's based upon sensory input at the
current moment
um cool i'm gonna just
move on to the next slide and just a
short little quote by buckminster fuller
because sometimes i just have to he
wrote rope may not be much like water
but not is like the wave and this is
from the synergetics framework
and it's getting at this notion that
there's a difference between structural
integrity
which is like the rope and the water
which are quite different they're both
static and stable
to pattern integrity and knots and waves
can travel
through a medium and so by virtue of
traveling through medium
kind of like that domino wave there's
something similar
about these traveling waves that goes
beyond the

mere structural integrity of their media
and so i just a thought question for
later is really
is active inference a structural
integrity or a pattern integrity
i would say that it's a pattern
integrity and just bringing up this
notion that many far from equilibrium
systems
though their mechanisms are extremely
different for example the bacteria
and the person there's going to be
similarities because
of the way that they're needing to
persist from a cybernetic perspective
enough there let's get to that question
about structural
aspects of attractors so here
a participant provided feedback and
wrote why can't
the recognition and generative models be
considered as structural representations
so they're coming from a structural
representationalist perspective
they wrote in which case the dynamics of
the structure are taken into account as
part of the representation
even if we're appealing to dynamical
systems the states of any particular
dynamical system
over time make up the structural
representation
take the lorenz attractor the lorenz
attractor has a specific approximate
structure around two attractor states
but the states and the trajectory of any
point are sensitive to initial
conditions

so any two initial points will lead to
different precise physical states and
dynamics
but will still travel through a very
similar trajectory which hovers around
these attractor basins
the states and the dynamics of a set of
points over time
can be taken as a whole and be
identified as embodying structural
representations of the lorenz
system so this is something from the
realm of chaos and complexity theory
and the lorenz attractor is an attractor
that describes a simplified system of
equations
describing the flow of fluid and it's a
fluid of uniform depth
and there are also parameters for the
imposed temperature difference
the gravity parameters the buoyancy the
thermal diffusivity and the kinematic
viscosity
and the simple physical analogy here is
like heating up a pot of water there's
heat coming from the bottom
and there's cooling coming from the top
and there's this churning happening and
it goes around these two different
attractor states
so in this case maybe alex uh
or anyone else who raises their hand
what is structure
and what is representation and if we
agree on what this system is then what
is this representation debate that's
happening
mel

mel unmuted mel
or maxwell go ahead all right okay let's
go to alex
yeah mel if you want to speak yourself
it's raised then we'll go to alex keefe
and they'll raise their hand first i
think yeah i was going to say i like
alex okay perfect
okay well i was going to say hey let me
talk
i don't have much to say about the
lorenz attractor um
it looks like a i mean what's i'm not
sure what's supposed to be represented
in this case
like it i have an intuition in the case
of we're dealing with like
a biological system because you know
we're
intuitively representing some kind of
environment i'm just not sure
i'm not sure how to begin with this
example
um right we've got the state space
and this nice graphical representation
of the states of this of the state space
but
um unless we interpret this entire
system as representing something i don't
know what to say about
its sort of content
okay cool mel
i don't know what a recommendation
is and so i
for that reason i guess i think that the
question of
what a representation is and
what in the world is a representation is

is maybe too ambitious only because i
haven't solved it
but there's a related question that's
less ambitious which is
what is the difference between a
biological
function and a representation
um and that i think is more manageable
in some respects um
in phil biomind cogside we
speak of the disjunction problem so we
speak of and i know
maxwell like hates this framing and
thinks it's updated and we don't need it
anymore
um the idea is that in order for
representation to be a representation it
needs to have the capacity to represent
and we have a similar idea in biological
function which is
that in order for a function to be a
function it needs to have
the capacity to malfunction right so you
need malfunction dab function you need
misrepresentation that representation
um but how do we differentiate
a function from a representation
good question maxwell then alex
cool um well so regarding the question
what is a representation
we targeted a very specific account of
representation
i mean in the philosophy of cognitive
science a representation is basically an
internal thing within an organism that
stands for
uh some feature of the environment or
the organism's own body

in a way that the organism can leverage
to you know do something interesting
we focus more specifically on the claim
that
generative models are structural
representations
so uh i mean this has
been an account that's been uh you know
opium o'brien first worked on this in
the mid-2000s
uh and then glazievsky and milkowski in
the context of predictive coding
and then generative models um i mean
yeah it i i think alex and i are gonna
still
disagree about this even after talking
about it for like dozens of hours at
this point but
um yeah the main
the main reason why uh in this paper we
argue that they're not
is that the generative model isn't
encoded in anything
uh and this is like um one of the main
differences between more traditional
bayesian
brain architectures and active inference
as dan was pointing out really early on
uh in traditional architectures well the
recognition and generative
models are just inverse the ones of the
others the the recognition model is a
mapping from the data to states
and uh the generative uh model is a
mapping from states to data
so it's really just the same kind of set
of connections and everything but it's
just like which direction are you

considering it

like from the top down flow or

from the from the uh the top down floor

or the bottom up flow of information

um

yeah well i mean that was a really a

nice explanation of what structural

representationalism is so

uh i don't have much to argue about

there um

and in general i don't i don't want to

argue more than is necessary

but you know that about me but um i

think um the

the reason i want to insist on some of

this stuff is just to for my own basic

sanity to just feel like i have an

understanding of what's going on with

this stuff

right um so the reasons despite i

understand

i think i understand exactly why from an

active inference perspective

um you wouldn't want to say that the

generative model is a

is encoded um let me just say first i

think

the question of whether or how it's

encoded is maybe a slightly distinct

question from whether it's a

representation and the reason i think

that it has to be a representation

well there are many reasons here one

argument the recognition model was an

approximation to the generative model

right

and so that's one example of like a

rational or probabilistic relationship

between these models and i don't think that that makes any sense if the generative model is not a representation also um so that's how that works under active inference i i agree that in in a more traditional bayesian brain sense the recognition model is trying to approximate uh the generative model i mean that's that's essentially how it's working um it's an approximate posterior approximately it's an approximate posterior but it isn't approximating the generative model well it's approximating the posterior under the generative model right so so right and both both the generative and the recognition models are concerned their distributions in part over uh i have to be careful here not the actual external states necessarily but over hidden hidden states i don't know how on earth you get those into the picture if it's not a representation that you're talking about because the hip states are environmental states they're not necessary right we have to be as you point out in the paper we have to be careful to distinguish the generative model from the generative process we can't just identify what actually happens with the generative model

we i think we need to have this is one
of the fundamental questions that i
think an activism
often is faced with is how do we deal
with yeah i mean mel brought up
misrepresentation
um so the fact that the fact the point
is
the way that you see the world the
organism or creature system sees the
world is not necessarily the way it is
but we still have that distribution over
external states as
essentially part of the generative model
let me just say one more thing here
about
encoding so the reason i think if you
look under the hood a little bit so
the generative model in under active
inference as i understand it
can roughly be identified with the
non-equilibrium steady state density
right
something like that um so
i i mean i would just say right the
dynamics
the dynamics sort of instantiate the
generative model i think that's a really
cool point and i actually think that
structural representationalists didn't
do quite enough to
emphasize the fact that the generative
model is used as a control system
um but i think if you look under the
hood there will be some features of the
system like stable
structural features in virtue of which
it has that nest density

um so in fact you will find some stable
thing that you could think of as
encoding
the generative model maybe at least
that's why i still have
you know i'm still attached to my views
on this cool
let's do alejandra and then mel
yeah um i think i i
i agree with alex or where are
i'm kind of confused how
the generative model can be conceived
um just like this
inactive process um
really i was like reading the paper
all over again and um
i actually don't get it um for me it was
um the recognition density is the
inversion of the generative model
so uh if it is not encoded
like anywhere it's
i i feel kind of uh lost
there um
for me yeah these top-down connections
this is the generality model
talking about the brain specifically and
the recognition density
is related with this bottom up
connection
so you can recognize um your best
guess right so if it is not encoded
um what can be said about these top-down
connections um
i don't know yeah maybe i can jump in
here and just
provide some points of clarification
sure that about the generation wall if
that's okay

yep um okay

so uh is screen sharing activated um

you could share your screen if you

yeah you probably should be able to

okay let me try this

okay so this is from a paper called uh

the graphical brain

and this is what the generative models

look like all right uh can everyone see

my screen

yep also we're gonna be returning to

this in the center um

yeah so the generative model is defined

as a joint probability density

over all of the variables of interest in

the system right so

uh in this case uh the variables of

interest are the states that we're

trying to infer which are here

s uh the policies that i'm trying to

pursue

which is here π and the data that i'm

observing at any given time step

which is here denoted o um

just so everyone uh is clear time flows

from left to right so this is the first

state this is the second state this is

the third state

and essentially uh the model you only

have access to this data point these are

counterfactual data points

uh and the whole thing is updated

basically at every time step

um and so look it the the generative

model itself like i said is just a joint

probability density it's like the

probability of all these variables

connected with ands effectively and

when we say that the generative model is not present in the system i mean that this density here this joint probability density is never encoded anywhere in the system uh because the generative model itself is factorized so uh basically you take this joint probability density and then using some uh you know bayes rule and other uh you know chain rule etc uh manipulations you can write this joint density as the product of a bunch of uh likelihoods and priors and it's these likelihoods and priors that are uh updated uh constantly as part of the recognition density so this this prior uh you know about your your data given your states and this other prior about your states given the next date and the policy and all this these are updated uh dynamically and their current value uh like all of them together can collectively comprise the generative model uh yeah that's right uh sorry the recognition model the uh generative model is how these quantities are all connected the ones to the others so like if you want to think about it heuristically uh you know this uh all of these specific parameters like the s the π the o

all that like they're all of those
values together
as they're being updated uh comprise the
recognition
density and the generative density
or the generative model is really just
these relations
between uh the different quantities
as they change so it's it's literally
that
the um the
inference process itself
is what holds all these quantities
together and the generative model is
just a description of how those
quantities flow together uh
yeah so again it's like the domino's uh
falling over
thing uh you know the the the wave of
dominoes only exists in that motion
and and similarly the generative model
only exists
in that kind of coordinated inference or
update dynamics
of the quantities that are part of the
recognition density
um yeah and the reason it's called the
generative model at all is just because
it's borrowed
it's a terminology borrowed from machine
learning uh
in machine learning um the the joint
probability density over all of your
variables is known as a generative model
so it's it's called that way
just because that's the that's the
technical
term uh it's it's also called that way

because if you're starting from a joint density over all of your variables you can actually generate fictive data that you would expect under this configuration

um of different uh parameters

oh thanks for that clarification maxwell will have those middles

we'll have mel and then we'll carry on with that sounds pretty good to me go for it mel

i think kate and mill have yeah yeah okay kate and then mel oh i just wanted to um

follow up to what maxwell was saying about structural representation can you hear me yep go for it can people hear me yes yes yes we can hear beautiful

awesome yeah just to follow up to what maxwell was saying about um structural representations uh the the sort of detachability of the representation right that's that's what should get us the distinction between um a biological function and uh at least a structural representation right because um my feet and legs and knees are um a representation of ground in some very minimal sense in the same way that fish fins are representations of of fish fins and bird wings or representations of fluid dynamics right um but this isn't detachable from immediate

environmental circumstance for offline
use

right um so that's what should get us
the difference between

uh function representation but i'm i'm
curious to see what role that plays

in in the ftp and active interest um

if any do we want to do we want to is

there is there merit and retaining a
distinction between

function eruption patients under under
active infants in the sap or are they

just

sort of continues with one another nice

very

good question kate did you have

something there

okay okay i'm gonna just continue with

the slides because i'm not sure

i think kate's up next yeah oh sorry i

wasn't first just wanted to apply um

i think i'm still with the

in the case of the the paper was really
useful for me in like

helping me understand what the

generation model was in the fep almost

all the way like when

it's described as like a statistical

description of the generative process

the way you've explained it here

all of that makes sense to me um

but then there's also parts in the paper

where it says

where is like the generative model sort

of

um is what the organism expects and it

guides what the organism does

which makes it sound like the generative

model is playing this kind of
causal role like not just the generative
model itself
um and i find that kind of confusing if
the given the generative model isn't
something encoded by the organism
um or by the system um
but i'm still i get the idea of it as a
statistical description of the dynamics
that
sounds i just don't understand how in
that
how if that's all it is that it's
something the organism uses it seems
like it's something we use
to describe the organism rather than
something organism itself is in any way
using um that's a good point i i've
proposed in a recent paper um
that came out in entropy not too long
ago that uh
both are correct so the you basically
the
the correct way to interpret the fep is
as instrumentalism next nested within
instrumentalism
meaning that we can have a model
theoretic
uh philosophy of science reading of the
fep
and you know for that matter any other
theory of cognition
as this is a model that we as scientists
are using to explain
the behavior of organisms
what i've been trying to argue is that
the fep itself at the theory level
is also saying the organism is

exploiting

the statistical structure of its body

and movement aka the generative model

to guide adaptive behavior so the model

really is

uh these harnessed relations between the

variables that have to

obtain for survival to persist

uh and you know so it's it's sort of

this idea that once the system kind of

gets

moving then there's kind of a an

inertial

dynamic thing kind of keeping it moving

so like you know my core body

temperature is 36 and a half degrees

and the fact that it is keeps a bunch of

other processes in play

that in turn keep me preserving my

temperature

you know by initiating uh adaptive

behavior like putting on a parka when it

gets cold and so on

uh so uh

yeah i mean both both kinds of

generative models are used

separately in dynamical causal modeling

uh of the kind that the fristen group

used to model the spread of covid we're

not assuming

that the process that we're modeling is

itself an active inference agent

um and it so you can do that

uh but you can also make the additional

assumption that it is

and then you're in the game of active

inference proper

cool thanks max oh go ahead kate

the thing the kind of way i get there
then it sort of seems
like you're losing the distinction
between the vehicle of the model and the
target of the model such that
i worry it sounds a bit like just saying
everything is a model of itself
essentially um i mean effectively that
is what we're saying
um and that's where it gets a little
hazy for me like you know
that's probably the point of
disagreement between alex and i also i
want to say the generative model just
is the organism in the sense that like
this
this specific you know statistical
structure
that harnesses all of these different
you know existential variables as
mel was saying like this really just is
the organism
like this kind of statistical structure
that keeps kind of
reiterating itself and coming
into existence that just is the organism
and that's one of the reasons
why i don't want to say that this model
is a representation of anything
you could just call it you know the
phenotype
we call it a generative model because
formally speaking
the same construct is borrowed from a
field where in which it's called a
generative model but
you can think of it as the organism's
phenotype and precisely because it's a

bit weird

to think of the model as modeling itself

i kind of resist

that interpretation and say well you

know if anything is a model really in

the more traditional sense it's this

recognition density that's performing

posterior state inference

and i'm sure that alex will tell me why

i'm wrong

in a second it's it's a great

response let's just move on through a

few more of the slides because there's a

lot in the figures that i think

gets at this um mel has their hand

though i'm sorry

go ahead i just want to i just wanted to

introduce it i think

i think maxwell is possibly mismodeling

himself i think he's

i think he's actually a tad bit more

realist

than he's letting on oh maxwell

a realist at heart on my bad days

exactly if you can't if you can't uh

take care of me

at my most real so in this next slide

one of the feedback just from a

participant just to give an

example and we won't spend too much time

here

but i like this idea that the beliefs

don't have to be verbal

uh that the physical therapy context

shows us how the range of movements that

the body does not

want to go to for example due to a

traumatic experience or injury

embodies a belief that doesn't involve language and this was a paper with uh liminowski and fristen from 2020 where a person's hand was opening and closing and then they were getting a vr visual that was delayed or was inaccurate and so there's so much here about how our sensory feedback our proprioceptive as well as our visual feedback multimodal systems are being used and that's the mess that would take us you know so many hours and more to clarify this is what the representation questions are about so it's like the person what is the representation of the current state of their hand being closed at that just pure phenotype level maybe somebody with a ruler says well it's whether the hand is closed or not but it's not just that because if you're getting visual input that's messing with you it changes your behavior so there's something else happening and that's really a lot of what we're talking about um i think going through the figures two three four are going to be really helpful before we have closing thoughts and thanks everyone for bearing with this was sort of a weirdly technically altered stream briefly the bayesian networks so bayesian networks are from bayesian statistical

sciences and in these diagrams
the nodes are variables they're like
random variables stochastic variables or
parameters
the edges are statistical relationships
they're probabilistic relationships and
they can be undirected
or directed these models are quite
common in statistics
and also there's some really nice
aspects about them
in that you can bring prior knowledge to
the table with your priors and you can
also estimate it directly from the data
with a parametric empirical base
and to give an example about what this
looks like here on the top right
we can see how like an accident
increases the likelihood that you'll
hear
sirens an accident also increases the
likelihood of
having a traffic jam bad weather can
increase the likelihood of accidents or
traffic jams
so these bayesian networks are pretty
commonly used in a lot of statistical
areas
and we can contrast that with what are
called forney factor graphs
and working through this devries and
first and paper was definitely
quite an experience this is a complex
paper and there's a lot in it
but in a forney factor graph as they
write in the paper it's a type of
graphical model that shares qualities
with bayesian networks and markov random

fields

for any factor graphs afford a visually
insightful representation of generative
models

especially beneficial for complex models
underlying hierarchical active inference
in a forney factor graph each factor is
represented by a node

and each variable by an edge so it's
like a flip of the bayesian

an edge attaches to a node if the edge
variable is an argument of the node

and so here on the right is a table from
the paper of devry's and fristen

and it shows how we can think about
these functions like addition

subtraction multiplication

as having variables coming in and out
and

just to really quickly summarize the
similarities and differences in the
bayesian networks the nodes are
variables

whereas in the forney graph the nodes
are the factors or the functions

in the bayesian networks the edges are
the probabilistic relationships amongst
variables

whereas in the forney factor graphs the
edges are the variables themselves

a cool trick about bayesian networks is
this model inversion

which is that the model can be gleaned
from the data

and also the data can be generated from
a model another cool trick of forney
graphs

is that they specify a specific order

for factorizable computation
making extremely massive models
tractable
and then a last cool fact about a
bayesian network is that's where the
markov blanket formalism comes from
which is it's the set of nodes that
insulates one part of the network from
each other
from a probabilistic perspective though
there's a there's a lot of nuance in how
it's used in the free energy principle
framework and the cool fact about the
forney graphs is that it's
linked to information theory and message
passing computation and other math
so in figure 2 we see
this generative model in active
inference and this is actually the model
that maxwell had
on his screen earlier and i annotated
all the pieces
so just as maxwell was saying time flows
from the left to the right
and in this is a again a bayesian
network representation
so the the boxes or the circles and
there's different types of them which
are described in the caption
the boxes in the circles are like
variables so the initial state is a
vector
the hidden states are variables the
state transition matrix is
how the hidden states map to each other
that's b a
is the map between the states and the
observations

and then figure three is the same thing
as a for any factor graph and so this is
showing that there's this equivalence
with representing something as a
bayesian net
and a forney factor graph and the forney
factor graph
allows for these extremely large scale
hierarchical models
to get factorized and then computed in a
tractable
way and so i should say like the
advantage of using a forney style factor
graph is that the message passing
is just you can just read it directly
off the thorny style factor graph
and there's an algorithmic way of moving
from a bayes net to a 40 style factor
graph
so typically intuitively the bayes net
is easier to write
because you can you know you can just
literally write down
the the dependencies between all of your
factors
like this uh so you know like i don't
this state depends on this other state
and you know there are these transitions
and etc blah blah
and so you know from this you can move
to the 40 style factor graph and that
tells you
just directly how that will be
implemented in
active inference i should say that the
reason why i
we discuss these at all is really just
to show in the next figure how the in

figure 4 how the generative model
and the generative process kind of link
up uh
i mean to me this is where the the
interesting stuff happens where
like literally the the generative
process
and this is a model of the generative
process right like how we think it's
structured
it's in this little box here um and then
the 40 style factor graph represents the
message passing
that is ongoing yeah outside the box
uh to keep all the the quantities within
you know phenotypic bounds
thanks maxwell and yeah i know that was
a little bit of a
rapid sprint through these figures but
it's so cool that there's a formal way
to relate the bayesian and the factor
graph
models and just as you specified it's
easy to specify sometimes the
bayesian network just like oh well the
traffic jam depends on the weather and
the weather depends on this other
feature
and then the factor graph allows us to
have
an order of computation and a really uh
implementable way to compute it
and also it has some similarities with
message passing for example in neural
network systems
so it's it's an interesting question
that i've been having is okay we talk
about this is as a bayesian

theory but if it can also be represented
in another way
is it is it really a uniquely bayesian
theory or what do we
gain by saying that it's a bayesian
theory it's uh within the domain of
bayesian inter-transformability uh or
interoperability
but that doesn't uniquely disambiguate
it and that's why i think some of these
philosophical questions that we're
talking about here like about
representation and about
cause are so open because
uh it's it's not uniquely specified by
the free energy principle or by active
inference like how exactly we should
answer these
philosophical questions because in the
end as maxwell said
it's the technical details this is what
we're talking about
when we're talking about the the free
energy minimization and how that bears
upon the policy selection
this is what's happening and also one
more point before if anyone
raises their hand to provide a comment
is um
notice that the free energy is linking
to the policy selection
 π p for policy and that means that
the organism isn't just minimizing the
free energy of the observations given a
sort of neutral representation
actually the policy representation
specifically how it bears
upon how states transition into each

other
that's what the actual computation is
being done
upon so the observations are being used
to run then
upstream through the hidden states
and their transitions into the policy
that ultimately dictates what's going to
be done by being selected by the
organism or the system as the most
likely so this is really how deeply
action is embedded in these models
it's not that there's a latent
representation of the external states
and then you're doing a best guess on
what you should do it's actually like
action is at the top here the policy is
at the top and so i think that's really
a nice
feature of these models and it becomes
graphically clear and then if somebody
wants to have philosophical framework a
or philosophical framework be on top of
this structure
or if they want to propose a different
structure for how observations and
policies are linked
that's something great it's something
tractable that we can talk about
but this also returns to the metaphor
discussion this is the real
thing as it is um cool i know we had to
sprint through there but it's just um
coming near the end of our time so any
thoughts on these
images or are people ready to sort of go
to our
closing thoughts i was just saying the

equations are more interesting but
yep the equations are interesting and
and the 2017 de verizon fristen and this
model it just reminds uh me that if we
can write down that bayesian
relationship the probabilistic
relationship between variables
then there's going to be a process model
that is going to be how we can link it
and uh these captions are very
informative and there's really a lot to
say here but enough for today
let's close it out um we can hear any
last thoughts but
while people are preparing their last
thoughts and raising their hand if
they'd like
i would just like to thank everyone for
participating we're learning
and developing each time figuring out
the tech each time
we provide follow-up forms to the live
participants so just check the calendar
invitation
and it would be really helpful to have
feedback we also
request feedback suggestions and
questions from
the participants live in the comments or
otherwise
and uh just stay in communication we
really appreciate everyone's engagement
here and it's been
such a great learning experience so if
anyone wants to raise their hand
alejandra first and then followed up by
anyone else who wants to
let's just have some closing thoughts

[Music]

well i was wondering if
if it's possible to like model updating
if
when you are doing offline cognition
um without without like an explicit
policy you are not doing
something online when this interaction
bodily interaction you are thinking
about something and then you have like
this
uh like you know said like this eureka
uh feeling
so this this is kind of a model updating
um without action
um i don't know the short answer is yes
i just linked the paper to it
we uh we've created a ruminating
agent um so basically this agent
combines the affective
inference uh scheme that we
uh have developed over the last two
years where you're effectively
inducing a new hierarchical layer of
state inference
based on state and precision estimates
at the lower level
and these are affective states that then
guide lower level precision estimation
so we combine that with what's called
sophisticated inference
which is the new generation of active
inference agents
um you know vanilla
formulation current vanilla um
you could see that in the generative
models that we just showed
they include uh beliefs about

counterfactual observations that we
would make as we propagate states into
the future

sophisticated inference is about
propagating your beliefs
about the observations that you would
make in counter-factual futures
and so by combining these two things we
have an agent that isn't acting
uh but that ends up ruminating um
and catastrophizing about
uh its own cognition um so
i just linked uh the paper in the chat
um

we we submitted it to the uh the
international workshop on active
inference
uh so the it's a little poster now but
we're we're
slowly expanding that into a full
manuscript but the the kind of summary
six pager

uh that just does what you were
suggesting
is uh this is it yeah
thank you um but this is kind of
confusing with the an active uh
view of cognition well this is why we're
not we're not like uh

classical and activists so uh yeah we
reject the equal partners principle
it's just not the case that uh the body
uh the brain and the cultural
like mealier are gonna be involved in
the same way

for every task so like for rumination
you know you could you could i guess you
know appeal to a history of interaction

which is you know the reason why agents
are able to entertain beliefs about
other agents at all
and so on and so forth but yeah you're
right there isn't much
in action here and that's fine i think
uh
cool alex kiefer and then anyone else
who wants to raise their hand
uh yeah so i guess in lieu of a proper
closing
statement i just wanted to point out
that maxwell has some really cool new
work on
how the toronto population sort of can
be seen as an acting
uh inference on their own level and that
this is one way you could see
uh structural representations being sort
of realized so
like that one thing we didn't have time
to get into um but i just wanted to say
in general like
i think um yeah there is i think
regardless of the precise philosophical
take that you have on all this stuff
like
it's cool to disagree on that level but
this is this this literature has
produced some really amazing
work and i think it's it's pushing
forward in a really interesting way
and um one one advantage of active
difference certainly is that
you can actually write down the
generative model and and sort of
transparently see how it works
and you don't get that out of just

unsupervised learning
cool any other thoughts
awesome well thanks so much for the
discussion we're gonna
um re-watch understand technologically
what did
didn't work and for next week provide
all of you
plus anyone else who wants to
participate with more of a specific spec
sheet for how
you can prepare and make sure that it
will be a really like seamless
experience
and as far as the timing goes as well
next week we will move on to actin
7 which is going to be about variational
ecology
which is a really awesome paper close to
my heart as a biologist
as well and we would really appreciate
anyone who wants to participate in
these conversations next tuesday at 7
a.m pacific or the following
tuesday both of these discussions will
be really great
and it's quite a complex paper so i'm
sure we'll have a lot to go
into again so once again thanks so much
everyone
for your participation really brought up
some awesome ideas and
i hope we conveyed to those within and
without of the field
that there's a lot of discussion ongoing
by no means are these questions resolved
and it's an opportunity for people to
bring their unique perspectives to the

table

so thank you all again for your

participation

really appreciate